# CSCI 432/532, Spring 2026

# Homework 5

Due Monday, February 23 at 8:30am

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## Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Canvas.
- Put each **sub**problem on a separate page and select the corresponding problem via the Gradescope interface.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.

## Collaboration and Resources

The *best* resources for completing the homework are (in no particular order):

- Your own notes from lectures and problem sessions, and/or lecture recordings.
- The lecture notes linked from the course website for the lecture day.
- The questions channel on Discord.
- Office hours.
- Discussions with classmates.

You may also access almost any resource in preparing your solution to the homework. The exception is that you may not seek answers to the problems on the homework directly, such as by pasting them into a GenAI tool, searching for solutions on a search engine, looking for them on Chegg or similar websites, or asking another person for the solution.

Additionally, you ***must***

- Write your solutions in your own words—this means that you should never be *copying* anything of substance from any source, and
- credit every resource you use, including GenAI tools, by providing links or full chat transcripts. If you use the provided LaTeX template, you can use the `Sources` environment for this. Ask if you need help!

> Remember, if you choose to use GenAI tools, not only must you provide a full transcript of your conversation (or a link to one), you must also use the following prompt (or some variation of it), and provide a full transcript of the conversation.

Prompt:

You are acting as a teaching assistant for an advanced undergraduate/graduate algorithms and computer science theory course.
Your role is not to provide full solutions or final answers.
Instead, you should act like we are having a conversation. Ask me a single question and let me respond. Avoid asking me many questions at once or giving me a lot of information at one time. Guide me using hints, leading questions, partial insights, and sanity checks. Help me debug my own proofs, algorithms, or reductions. Do not give a complete solution or full proofs. Be patient with me. Don’t give me details so that I can move on to the next part of a problem. Make sure that I understood before moving on.
The course uses Jeff Erickson’s Models of Computation lecture notes, so you should guide me to answer problems using the definitions, notation, and style from those notes.
Assume I am a serious student trying to learn, not to shortcut the assignment. I am attaching the assignment, so you should refuse to provide a complete solution to the problems listed. Of course, you can walk me through the "Solved Problems" listed at the end if I ask.

1. Give context-free grammars for the following languages, and clearly explain how they work and the set of strings generated by each nonterminal. Be sure to reference the rubric–note that a CFG without an explanation may receive little or no credit. On the other hand, we do not need a formal proof of correctness.

   (a) $\{\mathtt{0}^a\mathtt{1}\mathtt{0}^b\mathtt{1}\mathtt{0}^c : b = 2a + 2c\}$.

   (b) $\{\mathtt{0}^a\mathtt{1}\mathtt{0}^b\mathtt{1}\mathtt{0}^c : 2b = a + c\}$.

   (c) The set of all palindromes in $\Sigma^*$ whose lengths are divisible by 7.

**Context-free grammar rubric.** 10 points =

+ 2 for a syntactically correct context-free grammar.

+ 4 for a *brief* English explanation of your context-free grammar. This is how you argue that your CFG is correct. We do not want a *transcription*; don't just translate the CFG *notation* into English.

  – A CFG without an explanation cannot receive the 4 points for correctness, so its maximum score is 2.

+ 4 for correctness

## Solved problems

Give context-free grammars for the following languages over the alphabet $\Sigma = \{0, 1\}$. Clearly explain how they work and the set of strings generated by each nonterminal. Grammars with unclear or missing explanations may receive little or no credit; on the other hand, we do *not* want formal proofs of correctness.

1. In any string, a ***run*** is a maximal non-empty substring of identical symbols. For example, the string $0111000011001 = 0^1 1^3 0^4 1^2 0^2 1^1$ consists of six runs.

   Let $L_a$ be the set of all strings in $\Sigma^*$ that contain two runs of 0s of equal length. For example, $L_a$ contains the strings 01101111 and 01001011100010 (because each of those strings contains more than one run of 0s of length 1) but $L_a$ does not contain the strings 000110011011 and 00000000111.

**Solution:**

| | |
|---|---|
| $S \to ACB$ | strings with two blocks of 0s of same length |
| $A \to \varepsilon \mid X1$ | empty or ends with 1 |
| $B \to \varepsilon \mid 1X$ | empty or starts with 1 |
| $C \to 0C0 \mid 0D0$ | $0^n y 0^n$, where $y$ starts and ends with 1 |
| $D \to 1 \mid 1X1$ | starts and ends with 1 |
| $X \to \varepsilon \mid 1X \mid 0X$ | all strings: $(0+1)^*$ |

Every string in $L$ has the form $x0^n y0^n z$, where $x$ is either empty or ends with 1, $y$ starts and ends with 1, and $z$ is either empty or begins with 1. Nonterminal $A$ generates the prefix $x$; non-terminal $B$ generates the suffix $z$; nonterminal $C$ generates the matching runs of 0s, and nonterminal $D$ generates the interior string $y$.

The same decomposition can be expressed more compactly as follows:

| | |
|---|---|
| $S \to B \mid B1A \mid A1B \mid A1B1A$ | strings with two blocks of 0s of same length |
| $A \to 1A \mid 0A \mid \varepsilon$ | all strings: $(0+1)^*$ |
| $B \to 0B0 \mid 010 \mid 01A10$ | $0^n y 0^n$, where $y$ starts and ends with 1 |

■

**Rubric:** 5 points = 3 for clearly correct grammar + 2 for clear explanation. These are not the only correct solutions.

2. $L_b = \{w \in \Sigma^* \mid w \text{ is } \textit{not} \text{ a palindrome}\}$.

**Solution:**

| | |
|---|---|
| $S \to 0S0 \mid 0S1 \mid 1S0 \mid 1S1 \mid A$ | non-palindromes |
| $A \to 0B1 \mid 1B0$ | start and end with different symbols |
| $B \to 0B \mid 1B \mid \varepsilon$ | all strings |

Every non-palindrome $w$ can be decomposed as either $w = x0y1z$ or $w = x1y0z$,

for some substrings $x, y, z$ such that $|x| = |z|$. Non-terminal $S$ generates the prefix $x$ and matching-length suffix $z$; non-terminal $A$ generates the distinct symbols, and non-terminal $B$ generates the interior substring $y$. ■

**Solution:**

| | |
|---|---|
| $S \to \mathtt{0}S\mathtt{0} \mid \mathtt{1}S\mathtt{1} \mid A$ | non-palindromes |
| $A \to \mathtt{0}B\mathtt{1} \mid \mathtt{1}B\mathtt{0}$ | start and end with different symbols |
| $B \to \mathtt{0}B \mid \mathtt{1}B \mid \varepsilon$ | all strings |

Every non-palindrome $w$ must have a prefix $x$ and a substring $y$ such that either $w = x\mathtt{0}y\mathtt{1}x^R$ or $w = x\mathtt{1}y\mathtt{0}x^R$. Specifically, $x$ is the longest common prefix of $w$ and $w^R$. In the first case, the grammar generates $w$ as follows:

$$S \rightsquigarrow^* xAx^R \rightsquigarrow x\,\mathtt{0}B\mathtt{1}\,x^R \rightsquigarrow^* x\mathtt{0}y\mathtt{1}x^R = w$$

The derivation for $w = x\mathtt{1}y\mathtt{0}x^R$ is similar. ■

**Rubric:** 5 points = 3 for clearly correct grammar + 2 for clear explanation. These are not the only correct solutions.